

The schematic diagram illustrates the USB to I2C bridge circuit. The USB module (J1) is connected to a USB to I2C bridge IC (U12, USBLC6-25C6TP4). The bridge IC is powered by +5V and +3V3. It connects to an I2C bus (SDA, SCL) which is then connected to an 8-channel ADC (ADC0 and ADC1). The ADCs are powered by +5V and +3V3. The ADC outputs are connected to an 8-bit digital output (GPIO_EXT) which is then connected to an 8-bit digital output (GPIO_EXT).

The diagram shows two ADC modules, ADC0 and ADC1, connected to various signals. ADC0 has inputs AN0, AN1, AN2, and AN3. It has an output ALERT/RDYQ. It is connected to SCL and SDA. Its address ADDR2 is connected to 0x1001000. ADC1 has inputs AN4, AN5, AN6, and AN7. It has an output ALERT/RDYQ. It is connected to SCL and SDA. Its address ADDR2 is connected to +3Y3 and 0x1001001.

The schematic diagram illustrates the pinbot_protection.kicad_sch file, showing two identical protection ICs (TCA9535PWR) connected to a 16-channel GPIO. The diagram includes pin headers for GPIO0_OUT to GPIO15_OUT, power pins (+3V3, GND), and signal pins (SDA, SCL, INT). It also shows internal components like resistors (R64, R65, R19), capacitors (C7, C8), and a 10k pull-up resistor (R10).

Pin Headers:

- GPIO0_OUT to GPIO15_OUT:** 16 pins for digital output.
- Power:** +3V3 and GND pins for power supply.
- Signal:** SDA, SCL, and INT pins for I2C communication.

Internal Components:

- Resistors:** R64 (330R), R65 (330R), R19 (10k).
- Capacitors:** C7 (0.1uF 50V), C8 (0.1uF 50V).
- ICs:** TCA9535PWR (Protection IC), PCA9555PW (Buffer IC).

File: pinbot_protection.kicad_sch

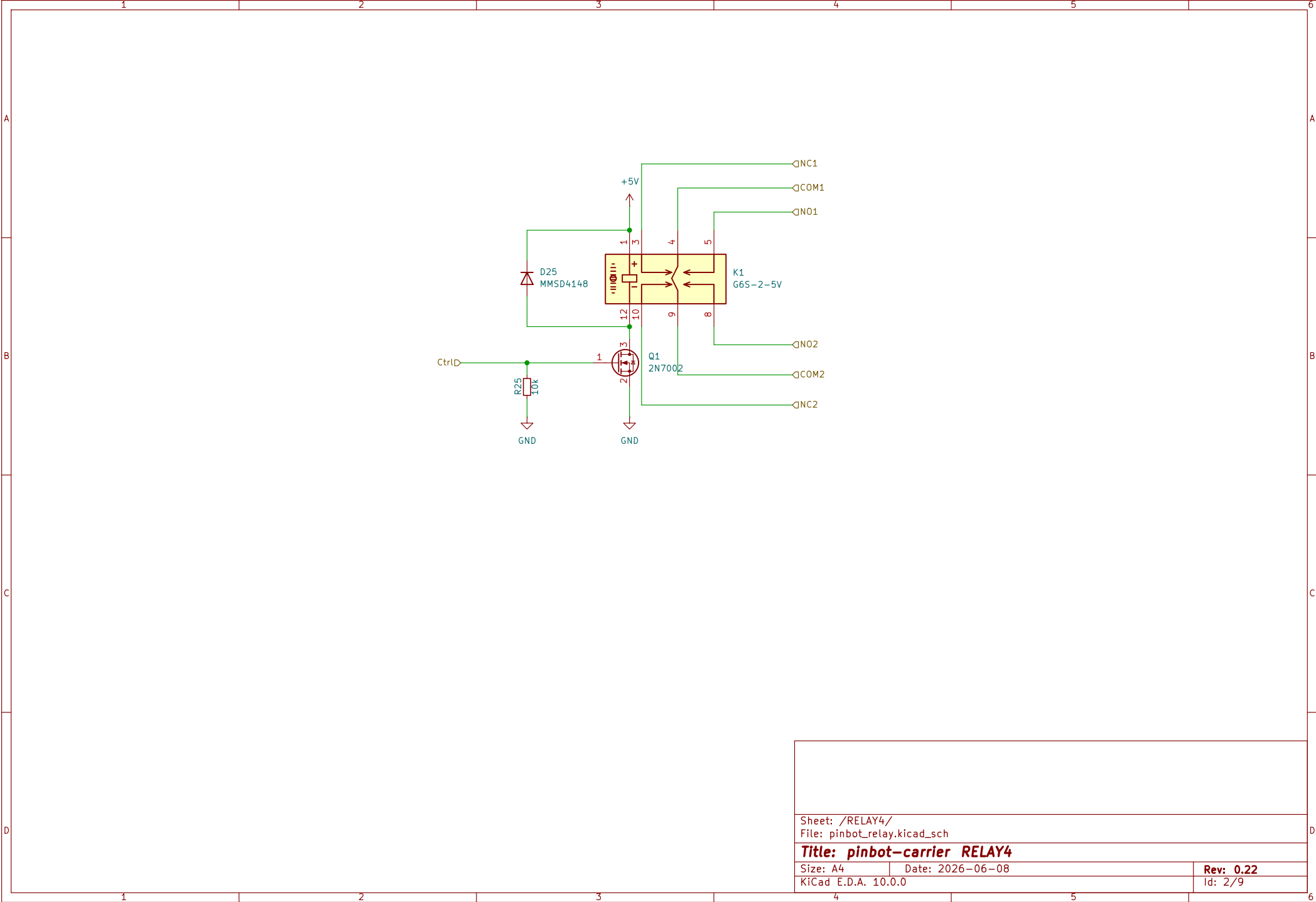
Pin connection diagram for the STM32F405VGT6 microcontroller. The diagram shows two 40-pin connectors, J2 and J4, with their respective pin numbers and functions. J2 is labeled "Conn_02x20_Odd_Even" and J4 is labeled "Conn_02x20_Odd_Even". The pins are numbered 1 to 40. The functions are listed next to the pin numbers. The diagram also shows the power supply connections: +5V_USB4, 40_D+, 40_D-, +5V, and +3V3. The microcontroller is shown in a 40-pin package with pins 1 to 40. The diagram is a schematic representation of the pin connections.

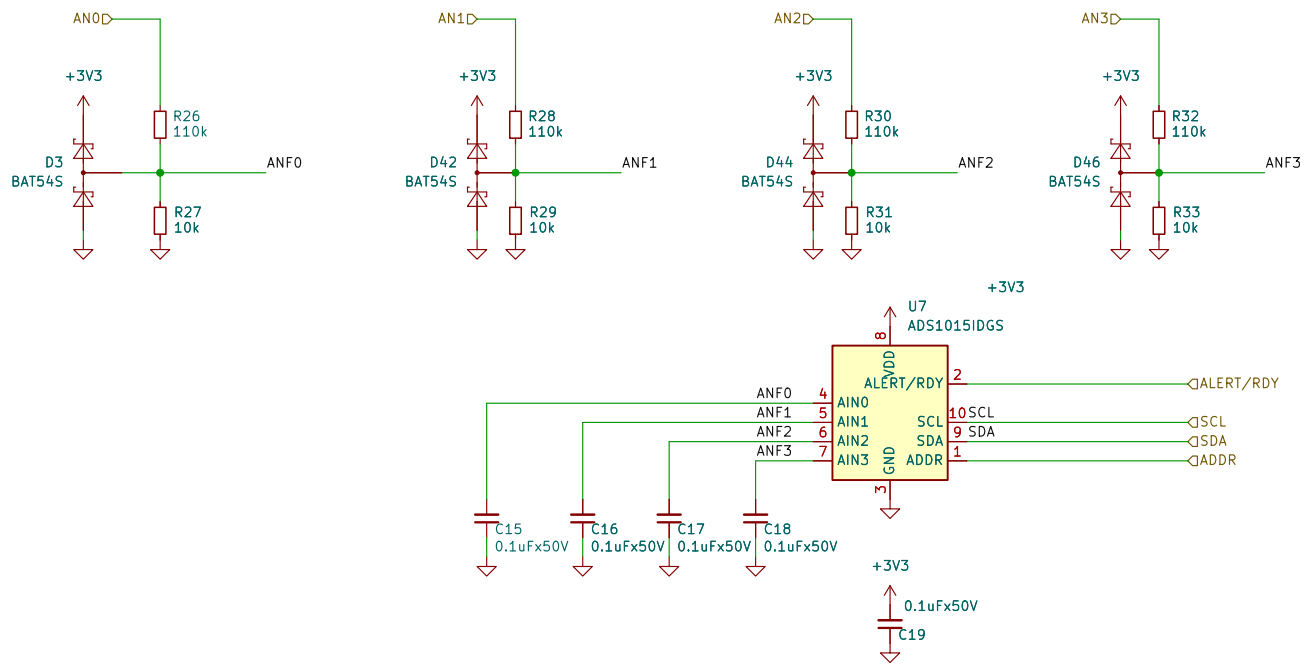
Pin	Function
1	AN0
2	AN1
3	AN3
4	AN5
5	AN6
6	AN7
7	
8	
9	
10	
11	GPIO0_OUT
12	GPIO1_OUT
13	GPIO3_OUT
14	GPIO5_OUT
15	GPIO6_OUT
16	GPIO7_OUT
17	
18	
19	
20	
21	GPIO8_OUT
22	GPIO9_OUT
23	GPIO10_OUT
24	GPIO11_OUT
25	GPIO12_OUT
26	GPIO13_OUT
27	GPIO14_OUT
28	GPIO15_OUT
29	
30	
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40	

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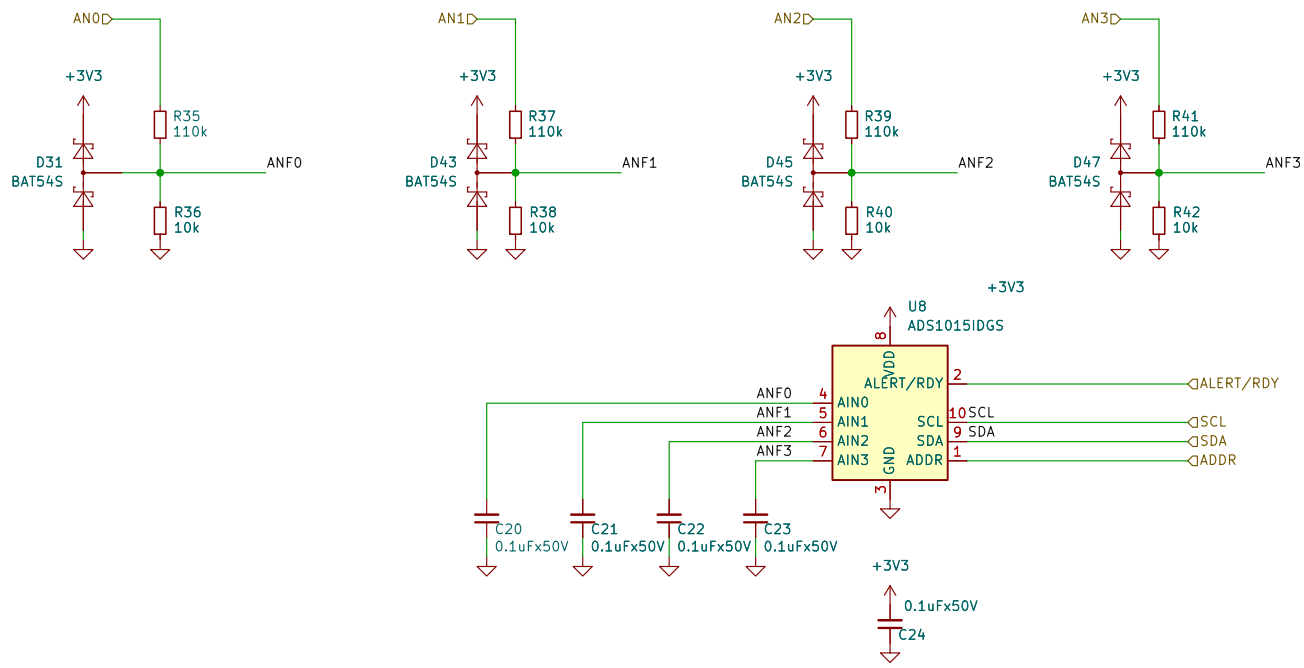
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-  H1 MountingHole
-  H2 MountingHole
-  H3 MountingHole
-  H4 MountingHole

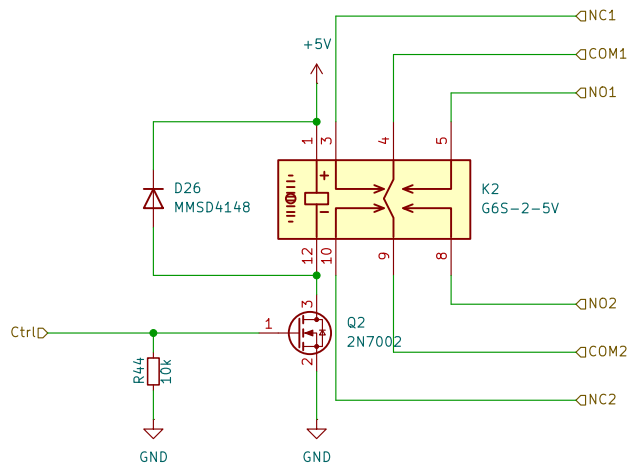




Divider 110k/10k scales input voltage by 12:1
(24 V → 2.0 V at ADS1015 input, PGA ±2.048 V)
0.1 μF capacitor at ADC input forms RC low-pass filter with divider output impedance
 $R_{eq} = 110k || 10k = 9.17\text{ k}\Omega$, $f_c \approx 174\text{ Hz}$



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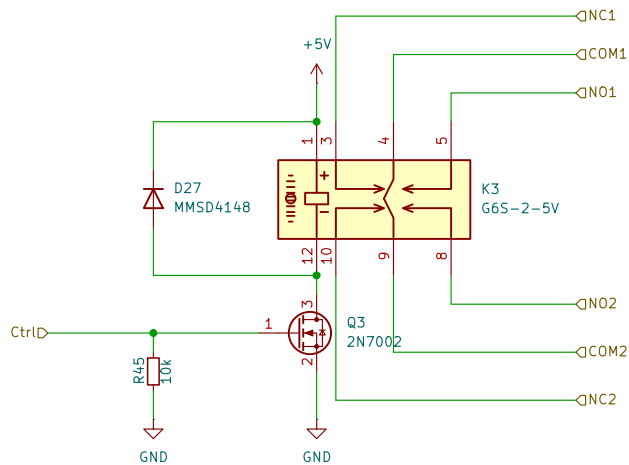
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Size: A4 Date: 2026-06-08

KiCad E.D.A. 10.0.0

Rev: 0.22

Id: 5/9



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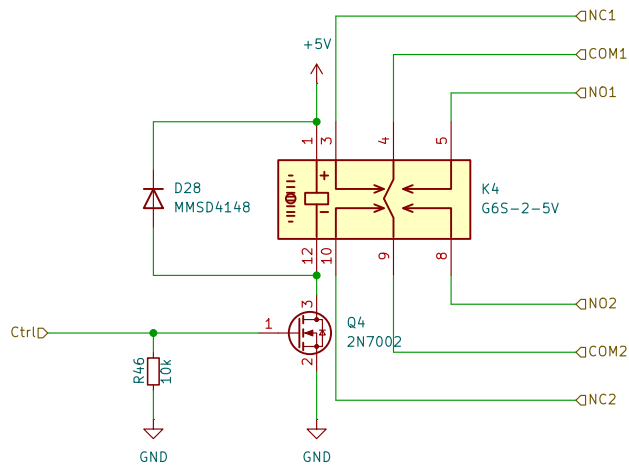
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Rev: 0.22

Id: 6/9



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Title: pinbot-carrier RELAY3		
Size: A4	Date: 2026-06-08	Rev: 0.22
KiCad E.D.A. 10.0.0		Id: 7/9

USB-hub with power control

